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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/093,039	01/04/2023	Bartlomiej Jan Pawlak	GF2022257-US-NP	8579

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Thompson Hine LLP
10050 Innovation Drive
Suite 400
Dayton, OH 45342-4934

EXAMINER	
MOONEY, MICHAEL P	

ART UNIT	PAPER NUMBER
2874	

NOTIFICATION DATE	DELIVERY MODE
12/19/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

ipdocket@thompsonhine.com

Office Action Summary	Application No. 18/093,039	Applicant(s) Pawlak et al.	
	Examiner MICHAEL P MOONEY	Art Unit 2874	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 2/24/23.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 1/4/23 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 1/4/23, 4/24/24.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

Claim(s) 1-2, 9, 11-14, 20 is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Kubby et al. (US 20040184710; "Kubby").

Regarding claim 1, Kubby discloses: a structure (variable optical attenuator 10) comprising:

• a semiconductor substrate (silicon substrate 60 in fig. 3; see ¶ 30);

• a first dielectric layer (buried oxide 56 in fig. 3, see ¶ 30) on the semiconductor substrate, the first dielectric layer including a first cavity (the free space between the two left-most portions 56 in fig. 3) and an edge (from ¶ 30 and figs. 1-3, 5, it is clear that the oxide layer is removed to the

left of the dashed line around optical slab 40, one edge being formed left to right near 58 in fig. 1) defining a boundary of the first cavity;

• an edge coupler including a waveguide core (the two waveguide cores 52, 54 on a cantilever that couple with stationary waveguides 42, 44), the waveguide core including a portion that extends past the edge of the first dielectric layer and overhangs the first cavity (the portions of the waveguide cores that are not fixed to the slab at 58); and

• a first heater (actuating beams 21 to be heated by current between ends 26 and 45, see ¶ 34) positioned adjacent to the portion of the waveguide core, the first heater spaced by a first gap (see fig. 2, see also ¶ 0031: thermal drive actuator 30 is connected to a tether 34 which comes near at least waveguide {WG} 54) from the portion of the waveguide core.

The same reasoning applies, mutatis mutandis, to independent method claim 20 the subject-matter of which corresponds to that of to claim 1, and which is therefore also not considered to be inventive.

The additional features recited in the following claims are also anticipated:

• Claims 2, 11, 12, 13, 14: Kubby (figs. 1, 3).

• Claim 9: Kubby (¶ 34).

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly

owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claim(s) 3-8, 10, 15-19 is/are rejected under 35 U.S.C. 103 as being unpatentable over Kubby et al. (US 20040184710; "Kubby") with obviousness evidenced by Peters et al. ["Electrothermal Actuators for Si02 Photonic MEMS", MICROMACHINES, vol.7, no.11, 7 November 2016 (2016-11-07), page200; "Peters"; already of record] and/or Sharma et al. ["Integrated 1x3 MEMS silicon nitride photonics switch", OPTICS EXPRESS, vol.30, no.12, 6 June 2022 (2022-06-06), page22200; "Sharma"] and/or Mahmoud et al. (US 11320718; "Mahmoud") and/or Takahashi (WO 2010/082524; already of record) and/or Davies et al. (US 20190369328; "Davies").

Regarding the additional features recited (beyond what has been addressed in the above 102 rejection):

• Claims 3-6, 15: Adding a second heater is an obvious modification, see e.g. Peters (two sets of heatable beams in fig. 2); or Sharma (two laterally spaced switching actuators in figs. 1, 2, 5); or Mahmoud (symmetrical actuating electrodes 106 in fig. 1A).

• claim 7-8: Coupling the waveguide to a laser device is obvious see for example Takahashi (fig. 8).

• claim 10: Adding a passivation dielectric over the heater and the waveguide is an obvious alternative in light of Davies (passivation dielectric 609 in fig. 6).

• Claim 16, 17: Obvious design alternative, see for example Sharma (fig. 5B).

• Claim 18: A taper is obvious in light of Sharma (fig. 2g).

• Claim 19: Kubby, ¶ 30, discloses the waveguides and heater to be made from monocrystalline silicon, making the heater from tungsten is an obvious alternative, see for example Takahashi (¶ 13).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Mr. Michael Mooney whose telephone number is 571-272-2422. The examiner can normally be reached during weekdays, M-F.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Uyen-Chau Le can be reached on 571-272-2397. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free).

/MICHAEL P MOONEY/
Primary Examiner, Art Unit 2874